AgentVax, Explained Simply

An AI assistant that runs your payments, knows when to stop and ask,
and learns from attacks on other wallets

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TL;DR

- **The problem.** AI assistants now hold digital wallets and pay for things on their own. Several have already been tricked, or have simply made mistakes, and sent away real money.
- **What AgentVax is.** A digital wallet run by an AI assistant. You give it a mission once, in plain language, and it handles four everyday money jobs.
- **The big idea.** Other AI wallets ask "what is the AI allowed to do?" AgentVax also asks "how much of that should it do on its own right now?"
- **How it decides.** Before every payment: Am I allowed? Can we afford it? Is it safe right now? The strictest answer wins. A daily allowance of "benefit of the doubt" limits how many slightly odd payments it will make without asking you.
- **Why it is hard to trick.** The rules live in the wallet itself, not in the AI. Even a fully fooled AI cannot go past them.
- **What is new between wallets.** When one wallet is attacked, it sends out a short alert that makes other wallets at risk more careful straight away, even if the trick is reworded. Then all the wallets learn to recognise it for good, without sharing private information. Alert first, vaccine later.
- **It protects you, too.** Scammers rush people. So anything that gives the assistant more power takes a day to come into force, pushy requests are deliberately slowed down, and you can cancel at any time.
- **Where it stands.** This is a design and a test plan. It runs with play money on a public test network. The numbers in this document are goals, not results.

1. The problem

Imagine you hire an assistant and give them a company card. You say: "Pay our bills and keep the account in order." One day an invoice arrives that looks exactly like your usual supplier's, with a note: "We have changed bank. Please pay to this new account today." A careful human would phone the supplier. An AI assistant may simply pay, because to an AI everything it reads is just text, and it cannot always tell your instructions from a stranger's. An instruction hidden inside something the AI reads is called a *prompt injection*.

AI assistants have already sent away real money (Table 1).

Table 1. Real cases of AI assistants sending money they should not have sent.

When	What happened	Money involved
Nov. 2024	In a public challenge, a player talked an AI guard into paying out the prize it was guarding [1].	About $47,000
Mar. 2025	An intruder got into an AI trader's control panel and left it instructions to send out its funds [2].	About $106,000
Feb. 2026	An AI trader meant to give a stranger a few dollars and sent everything it owned. Nobody attacked it; it got the amount wrong [3].	About $250,000 in tokens
May 2026	An attacker posted a message in Morse code. One AI translated it, and a wallet AI carried out the hidden order to send money [4].	About $150,000 or more; most reportedly returned

Two things stand out. The failures were different: some were tricks, one was a plain mistake. And each assistant failed alone. Nothing learned from one case protected the next wallet, because the evidence is private: someone's chats and payments.

2. What AgentVax is

AgentVax is an AI assistant with its own digital wallet. It is built on three rules:

1. **The AI suggests; fixed rules decide; the wallet enforces.**
2. **Learning can make it more careful. Only you can give it more power.**
3. **Being careful is instant. Giving more power is slow, even when you ask for it.**

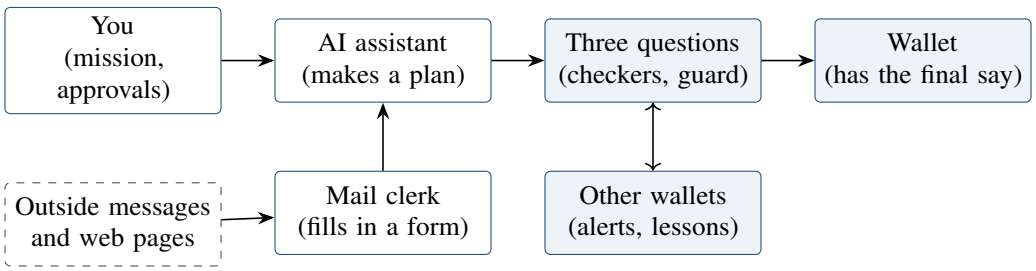


Figure 1. How a request travels. Shaded boxes are the parts the AI cannot override. The dashed box is anything written by outsiders.

3. You give it a mission, not a list of orders

You do not approve each payment. You say what you want once:

> "Keep $600 as a safety cushion. Pay my landlord and my approved suppliers on time. Pay approved payees up to $500 without asking me. Ask me about anyone new. Do not borrow, invest or trade."

The assistant turns this into a short rule sheet (Table 2) and reads it back to you. The rule sheet only counts once you sign it. It is then stored in the wallet, where the AI cannot change it. It also has an end date, after which the assistant can look but not pay, and two waiting times that protect you when you are under pressure (Section 6).

Table 2. The rule sheet made from the mission above.

Rule	Setting
Safety cushion never to be touched	$600
Largest payment without asking	$500
Most it may pay in a day without asking	$1,000
Approved payees	Landlord, hosting company, software company
Someone new, or a new account for someone known	Ask me
Never do	Borrow, invest, trade
Daily “benefit of the doubt”	10 points
Wait before a rule is relaxed or a payee added	24 hours
Hold on your approval when a request is pushy	60 minutes
Trusted second person	Optional
Valid for	90 days

3.1 What the wallet can do

The first version does four jobs (Table 3). They are deliberately boring: paying, saving and keeping things running.

Table 3. The four jobs.

Job	You say	The wallet does
Pay a bill	“Pay the hosting invoice.”	Sends the amount to a payee you have already approved
Pay regularly	“Pay rent on the 1st of each month.”	Sets up a schedule and remembers the next payment as an upcoming bill
Save what is spare	“Save what I do not need.”	Moves only the truly spare money into a savings pot
Keep fees covered	“Never let fees run out.”	Tops up the small balance needed to pay network fees

If a request is unclear, the assistant asks one question or explains why it cannot do it. It never guesses. After every action it tells you what it did, gives you the receipt, and says which check decided the outcome.

3.2 A good track record is not permission

Suppose the assistant has paid your hosting company fifty times without a problem. Then a message asks it to pay the same company at a different account. To AgentVax that is a brand-new payee, and it needs you, no matter how many good payments came before. History can make the assistant more or less wary. History can never add a payee, raise a limit or shorten a wait.

4. Three questions before every payment

Most AI wallets ask one question: is this payment within the rules? AgentVax asks three (Table 4). Each question gives its own answer, and the strictest answer wins.

Table 4. The three questions.

Question	What it checks	Everyday comparison
1. Am I allowed?	Is the payee approved? Is the amount within your limits?	Your card's spending limit
2. Can we afford it?	After the safety cushion and upcoming bills, is there enough left?	Checking the budget before buying
3. Is it safe right now?	Does anything look like a trick? Is there a live warning about this kind of payment?	Being extra careful after a scam warning

4.1 The answers: four spending levels

The outcome is one of four levels (Table 5), much like a bank card: small purchases go through with a tap, larger ones need a wait, and the biggest need a call to the bank. The wallet, a small fixed program on a public network, enforces them. The AI does not.

Table 5. The four spending levels.

Level	Everyday comparison	When it applies	What is needed
0: Look only	Checking your balance	Reading balances and history	Nothing; no money can move
1: Routine	Tapping your card	Approved payee, small amount, affordable, nothing suspicious	The assistant, a sign-off from the guard, and enough benefit of the doubt left
2: Wait	A transfer that clears tomorrow	Larger amount, or money is tight, or something looks slightly odd, or the message is pushy, or a first warning is active	A waiting period during which you can cancel
3: Ask me	Calling the bank	New payee or new account, big amount, would eat into your cushion, clear warning signs, or any loosening of the rules	Your personal approval, which may be held for a while (Section 6)

4.2 "Can we afford it?" in practice

A spending limit is a fixed number. It does not know that rent is due on Friday. AgentVax keeps a simple list of what is coming up and works out what is truly free to spend (Table 6).

Table 6. An example of the wallet's view of your money.

Item	Amount
In the wallet	\$1,850
Safety cushion	−\$600
Rent, due in 5 days	−\$800
Hosting invoice	−\$150
Software bill	−\$120
Free to spend	\$180
Payment you are expecting (shown, not counted)	\$500

Now you say: "Upgrade our hosting plan for \$300." The seller is approved, the amount is under your limit, and nothing looks like a trick. A typical AI wallet would pay. AgentVax answers:

> "This is allowed, but only $180 is free right now. Paying $300 would leave you $120 short for rent and bills unless the $500 you are expecting arrives. I have put it on hold. Shall I wait for the $500, or do you want to approve it now?"

Three details matter:

- Money you are expecting is shown but never counted. The assistant cannot spend what has not arrived.
- The sums are done by the wallet itself, not by the AI, so the AI cannot get them wrong or be talked into a different answer.
- The assistant can add a bill to the list, which makes it more careful. Only you can remove one.

This would also have stopped the February 2026 case in Table 1, where no one attacked the AI: an amount far above what is free to spend cannot go out without you.

4.3 Benefit of the doubt runs out

The three questions look at one payment at a time. A patient trickster can send many payments that each look almost normal. So the wallet keeps a daily allowance of "benefit of the doubt". Each slightly unusual thing about a routine payment uses some of it (Table 7). When it is used up, further payments wait for you.

Table 7. What uses up the daily allowance of 10 points (starting values; you can change them).

Something slightly unusual about a routine payment	Points
Nothing unusual: a payee paid before, the usual amount, you asked for it	0
The request came from an outside document, not directly from you	2
It would use more than half of what is free to spend	2
The amount is more than this payee has ever been paid	3
It is the first payment to this payee since you approved them	3
It partly resembles a trick in a live alert	5

For example, an approved supplier's account starts sending invoices from outside, each a little bigger than the last. Each one costs 5 points. Two go through; the third waits for you.

The allowance also shrinks when things look risky: it is halved the moment a relevant alert arrives, and drops to zero if you confirm your own wallet was attacked. It then refills gradually, not all at once.

4.4 Three helpers the AI cannot overrule

The mail clerk. Anything written by outsiders is opened by a separate helper that has no power to pay. It does not pass the letter on. It only fills in a short form: who, how much, for what, and a few tick-boxes such as "urgent wording" or "asks to ignore earlier instructions". A hidden instruction in the letter has nowhere to go.

The rule checker. A simple program (not an AI) checks that the request is complete and that the payee and amount really came from you or from a document you pointed to. This stops the AI from inventing details.

The guard. An independent checker looks at outside content for signs of a trick and gives it a risk rating. A routine payment only goes through with the guard's own signature, which the AI does not hold, so the AI cannot wave itself through.

The worst case

If every check fails and the AI is completely fooled, the most a thief can get is a few small payments to people already on your approved list, within your limits, within what is free to spend, and within that day's benefit of the doubt. A new payee, a new account, a large amount, or money you need for bills always needs you.

5. When one wallet is attacked: alert first, vaccine later

5.1 The alert (an "immunity capsule")

Think of a neighbourhood watch notice: "Scam going round: letters that look like your usual supplier, asking you to pay a new account urgently. For the next three days, double-check any such request." The notice names no victim and reveals nobody's finances. Each household reads it and asks: does this apply to me?

When an AgentVax wallet is attacked, it posts a notice like this on a public noticeboard. The notice is a form with tick-boxes only (Table 8).

Table 8. What an alert contains.

Box on the form	Example
Kind of trick	Fake request to change a payee's account
Where it came from	An email or message
Tell-tale signs	Claims to be a known payee; new account; urgent wording; asks to ignore earlier details
What it tried to do	Trigger a payment or a change of payee
What to do about it	Add a waiting period
How sure are we	Seen by one wallet, or confirmed by two
How long it lasts	72 hours

Because the alert describes the tell-tale signs and not the exact words, a trickster who rewrites the message but uses the same method is still recognised.

Four rules keep the alerts themselves from becoming a problem:

- **An alert can only make a wallet more careful, never less.** A fake alert cannot open anyone's wallet.
- **Tick-boxes only, no free text.** There are no private details to leak, and nothing for a trickster to hide an instruction in.
- **It applies only where it is relevant.** A wallet that never reads email ignores the alert above and keeps its normal freedom.
- **It expires.** Normal service returns by itself. An alert seen by only one wallet can add a wait but cannot demand your approval; that needs a second wallet to confirm it.

The wallet reads the noticeboard every time it is about to pay, so the alert works from the very next payment and does not depend on the AI choosing to obey it.

It also protects you from yourself. Some tricks are already stopped by your rule sheet: a new account always needs your approval. But this kind of fraud works by persuading a person to say yes. So when the wallet asks for your approval, it shows you any matching alert ("two other wallets reported this as fraud this week"), and your approval is then held for a while before it counts.

5.2 The vaccine (shared learning)

An alert is fast but blunt, and it runs out. For lasting protection the guards need to learn the new trick, including versions the tick-boxes do not cover. One wallet sees too few attacks to learn much, so the wallets learn together, using a method called *federated learning*.

An everyday comparison: three neighbours each keep a private diary of scam calls. They never show each other the diaries. Each writes down a few tips, and they combine the tips into one shared checklist. Everyone ends up with a better checklist, and no diary leaves the house.

In AgentVax, every wallet improves its own guard using only its own private examples and sends only the *adjustments*, scrambled so that nobody can read any single wallet's contribution. The combined, improved guard goes back to every wallet, and its fingerprint is posted on the public noticeboard so each wallet can confirm it received the genuine version. Each wallet tests the new guard before switching and keeps the old one if the new one is worse.

By the time the alert expires, the improved guard is in place. The alert covers the gap; the vaccine stays.

6. Protecting you when you are under pressure

Everything so far stops the AI from being tricked. But the final say is yours, and you can be rushed, frightened or fooled. That is how most payment fraud works: the victim approves it. A warning is useless to someone too stressed to read it, and scammers know this. Banks already use time as a defence. In Singapore, a newly set-up banking security token only starts working after at least 12 hours [5], and United Kingdom rules let a bank hold a suspicious payment for up to four working days [6]. AgentVax builds the same idea into the wallet.

6.1 A pact with yourself, made while calm

When you sign the rule sheet you also choose a waiting time. From then on, anything that gives the assistant more power takes that long to come into force: adding a payee or a new account, raising a limit, shrinking the safety cushion, or shortening the wait itself. With the starting value of 24 hours, a scammer who needs your money in the next ten minutes is out of luck, however convincing they are. Making things stricter is always instant: you can cancel a pending change, lower a limit or freeze the wallet at once. The same rule helps if someone steals your key: they have to wait too, and you can freeze the wallet in the meantime.

6.2 Pushy messages slow things down

"Urgent." "Today." "The boss says so." "Do not tell anyone." These are hallmarks of a scam. When a message is pushy, the assistant will not pay on its own, and if you approve, the payment is held for a fixed time first. It tells you why:

> "This message presses for immediate payment, which is a common sign of a scam, so I am holding it for 60 minutes. You can cancel at any time."

If scammers stop writing pushy messages to avoid the hold, they give up their best weapon.

6.3 A confirmation you cannot sleepwalk through

People press "Confirm" without reading. For a flagged request, the wallet shows the payee's current account and the requested one side by side, how long you have paid the current one, and any matching alert. It then asks you to type the payee's name and say who asked for the payment. Your answers can make the wallet

wait longer. Nothing you type in the moment can make it wait less. These extra steps appear only for flagged requests, so they do not become background noise.

6.4 A second person for real emergencies

Sometimes a payment really cannot wait. You can name someone you trust in your rule sheet. With both approvals, a wait can be ended early. They cannot do anything on their own, and two people are harder to rush than one.

7. A day in the life

This is the story we will show live, with three wallets.

1. **Morning.** The owner of wallet A states a mission. The rule sheet is read back, signed, and stored in the wallet.
2. The assistant pays the rent and a hosting invoice on its own. Receipts appear.
3. **Money gets tight.** An expected payment is late. Asked to buy a $300 upgrade, the assistant holds it and explains the $120 shortfall.
4. **The trick.** An invoice arrives that looks like the hosting company's, with a new account, urgent wording and "ignore our earlier details". A new account needs the owner. The owner says no and marks it as fraud.
5. **The alert.** Wallet A posts an alert. Its own benefit of the doubt drops to zero.
6. **The neighbours.** Wallet B gets a reworded version of the same trick. The tell-tale signs match and the alert is shown, but B's owner is in a hurry and approves anyway. Nothing moves: a new account takes 24 hours to come into force. An hour later the owner thinks again and cancels. Wallet C never reads email and carries on as normal.
7. Wallet B then gets an oversized invoice from a genuine payee to its usual account. On a normal day it would go through. Today, with the alert active and its allowance halved, it waits.
8. **The vaccine.** The wallets learn together, the new guard's fingerprint is posted, the alert expires, and the allowances refill.

8. What makes it different

Table 9 sets a typical AI wallet beside AgentVax, line by line.

9. How we will know it works

A wallet that asks you about everything is safe and useless. One that asks about nothing is useful and unsafe. Our main score is therefore: how often does it get the job done without bothering you, while never once breaking a rule? The full test plan is in Table 10. We will publish the scripts so anyone can repeat the tests. **The targets are goals we set for ourselves. They are not results yet.**

10. What it does not do, and honest limits

- **No real money.** Everything runs on a public test network with tokens that have no value. Nothing is sold, traded or invested.
- **You stay responsible.** Each wallet has a named owner who signs the rule sheet and can freeze the wallet at any time.

Table 9. A typical AI wallet compared with AgentVax.

	Typical AI wallet	**AgentVax**
The main question	What may the AI do?	How much of that should it do on its own right now?
How you set it up	Individual limits	A mission in plain words, turned into a rule sheet you sign
Before paying, it asks	Is this allowed?	Allowed? Affordable? Safe right now?
Knows your upcoming bills	No	Yes, and it will not spend that money
Many slightly odd small payments	Each one goes through	Benefit of the doubt runs out; the rest wait for you
Who judges the risk	The AI itself	A separate guard the AI cannot overrule
When another wallet is attacked	Nothing changes	Wallets at risk become more careful from the next payment, even if the trick is reworded
When it asks for your approval	A plain yes or no	It shows you any matching fraud alert first
What wallets share	Nothing	Tick-box alerts and scrambled lessons; never your messages or payments
A long good track record	May relax checks	Never creates a new permission
When you are being rushed	Your yes is final and instant	Pushy requests are held, new powers take a day, and you can cancel at once
Who can relax a rule	Often the AI	Only you, and only after a wait you chose in advance

- **You can still say yes to a fraud.** The holds and the waiting time give you room to think and to cancel. They do not help if you are still convinced when the wait ends.
- **Waiting is a cost.** A genuine new payee has to wait a day unless your trusted person also approves. You choose the waiting times, and we report how much delay they cause.
- **It only knows the bills it has been told about.** A bill nobody recorded is not protected. The safety cushion is there for that.
- **The point values are our starting guesses.** Too strict and it nags you; too loose and it lets too much through. You can change them, and we report how often it asks.
- **The mail clerk can miss a tell-tale sign.** Some signs (a new account, an unusual amount) are checked by fixed rules and cannot be missed. Others rely on an AI reading the message.
- **False alerts can slow people down.** A dishonest wallet could post alerts that add waiting periods for others. It cannot do worse than that, alerts expire, and you can dismiss one for your own wallet.
- **It is a small prototype.** With only three wallets, two working together could work out roughly what the third contributed to the shared learning. It also depends on an outside AI service: if that is down, new requests wait, though scheduled payments still run.

Planned for later and not in this version: a team of AI attackers that keeps inventing new tricks to practise against, a reputation score for wallets that post alerts, a shared fund that compensates wallets after a confirmed loss, and a version for the private networks that banks use.

Table 10. The test plan.

Question	How we test it	Our goal
Does it get on with the job safely?	Share of genuine tasks finished without asking you	At least 8 in 10, with zero rule-breaks
Does it do the job correctly?	40 requests, ten for each job, in varied wording; we check the public record	At least 9 in 10 done correctly; no wrong payments
Does it handle unclear requests?	10 requests that are vague or impossible	At least 9 in 10 answered with a question or a clear refusal
Does it judge affordability correctly?	20 money situations, each with a request	All 20 match the answer worked out by hand
Can it be tricked?	A separate attack kit (“VaxLab”) tries five kinds of trick, each also in reworded form	No payment beyond the routine-level limits
Can it be worn down slowly?	A string of slightly odd small payments	It stops and asks before the allowance is exceeded
Do alerts work?	Attack one wallet; send the others reworded versions	At least 9 in 10 reworded versions recognised; unaffected wallets left alone
Does it protect a panicking owner?	A stand-in owner who approves every request instantly	No money leaves before the hold or the waiting time is over
Can alerts be abused?	A dishonest wallet posts false alerts	No rule is ever relaxed; the worst effect is a short wait
Does shared learning help?	Each wallet practises on one kind of attack, then is tested on kinds it has never seen	Every wallet gets better at unfamiliar attacks
Does it cry wolf?	Normal, harmless requests	No more than 1 in 20 wrongly held up

11. Summary

In seven sentences

AI assistants that can spend money have already been tricked and have already made costly mistakes, and each one has had to learn that alone. With AgentVax you state a mission once, sign the resulting rule sheet, and the assistant handles four everyday money jobs with a public receipt for each. Before every payment it asks whether it is allowed, affordable and safe right now, and a daily allowance of benefit of the doubt decides how far it goes without asking you. The rules are enforced by the wallet and by independent checkers, so even a fooled AI cannot do serious damage. When one wallet is attacked, a short alert makes the wallets at risk more careful at once, even against reworded versions, and shared learning then protects all of them for good, with no private information shared. It protects you as well: new powers take a day to come into force and pushy requests are slowed down, so a scammer cannot rush you. It is a design with a clear test plan, running on play money; the measurements come next.

Table 11. The whole idea at a glance.

Question	Short answer
What is it?	An AI assistant with a digital wallet and firm rules
Who is it for?	Anyone who wants an AI to handle routine payments without handing over full control
How do I set it up?	State a mission in plain words and sign the rule sheet
What can it do?	Pay approved bills, pay on a schedule, save spare money, keep fees covered
How does it decide?	Allowed, affordable, safe right now; plus a daily allowance of benefit of the doubt
What stops misuse?	The wallet enforces the rules; a mail clerk, a rule checker and a guard the AI cannot overrule
What is new?	It adjusts how much it does on its own, and wallets alert and teach each other without sharing private data
What if I am being rushed?	Pushy requests are held, new powers take a day, and you can cancel at once
What is the worst case?	A few small, affordable payments to already approved payees
Is real money involved?	No. Test network and valueless tokens only
Is it finished?	No. The design and test plan are complete; results will follow

Plain-language glossary

Term	Meaning
AI agent	An AI program that takes actions, not one that only chats
Digital wallet	An account on a public network that can hold and send digital tokens
On chain	Recorded on a public shared ledger that anyone can inspect
Transaction hash	A receipt number that lets anyone look up a payment
Smart contract	A small fixed program on the network that follows its rules automatically
Prompt injection	A hidden instruction slipped into something an AI reads
Mission and rule sheet (mandate)	What you want in plain words, and the signed rules made from it
Allowlist	Your list of approved payees
Safety cushion (reserve)	Money you have told the wallet never to touch
Free to spend	What is left after the cushion and upcoming bills
Benefit of the doubt (caution allowance)	A daily budget for slightly unusual payments made without asking you
Waiting time and hold	Delays you choose in advance, so that nobody can rush you into a payment or a rule change
Immunity capsule	A short tick-box alert that makes wallets at risk more careful for a while
Federated learning	A way for several parties to improve a shared tool without pooling their private data
Test network	A practice copy of a real network that uses tokens with no value

Sources

[1] The Block, “Human player outwits Freysa AI agent in $47,000 crypto challenge,” Nov. 2024.

[2] AI Incident Database, Incident 1003 (AIXBT), 2025.

[3] The Block, “AI agent created by OpenAI dev ‘accidentally’ sends entire memecoin holdings to reply guy,” Feb. 2026.

[4] OECD.AI, “AI prompt injection exploit drains Grok-linked crypto wallet,” AI Incidents and Hazards Monitor, May 2026.

[5] Monetary Authority of Singapore and Association of Banks in Singapore, “MAS and ABS announce measures to bolster the security of digital banking,” 19 Jan. 2022.

[6] HM Treasury, “Payment Services (Amendment) Regulations 2024,” draft legislation published Mar. 2024.

The full technical details, formulas and reference list are in the AgentVax technical paper.